

Advanced Data Structures & Algorithms Program

16 Days • 6 Hours Per Day • Interview Focused Intensive Program

Day 1 – Linked List Fundamentals

Duration	Topics Covered
2 Hours	Singly Linked List, Insert/Delete, Traversal
2 Hours	Two Pointer Technique + Fast & Slow Pointer
1 Hour	Reverse Linked List Variations
1 Hour	Dry Runs + Complexity Analysis

Problems Solved in Class:

- Reverse Linked List
- Middle of Linked List
- Linked List Cycle
- Palindrome Linked List

Day 2 – Advanced Linked Lists

Duration	Topics Covered
2 Hours	Doubly & Circular Linked Lists
2 Hours	Merge Techniques + Intersection Problems
1 Hour	Partition & Rotation Problems
1 Hour	Contest Style Questions

Problems Solved in Class:

- Merge Two Sorted Lists
- Rotate List
- Intersection of Two Linked Lists
- Remove Nth Node From End

Day 3 – Linked List Interview Problems

Duration	Topics Covered
2 Hours	Reverse Nodes in K Groups
2 Hours	LRU Cache Design
1 Hour	Copy Random Pointer
1 Hour	Optimization + Discussion

Problems Solved in Class:

- LRU Cache
- Copy List with Random Pointer
- Flatten Multilevel DLL
- Reorder List

Day 4 – Trees Fundamentals

Duration	Topics Covered
2 Hours	Binary Tree Traversals
2 Hours	Recursive Tree Problems
1 Hour	Level Order Traversal
1 Hour	Problem Solving Session

Problems Solved in Class:

- Binary Tree Inorder Traversal
- Maximum Depth of Binary Tree
- Same Tree
- Level Order Traversal

Day 5 – Advanced Trees

Duration	Topics Covered
2 Hours	Diameter + Balanced Trees
2 Hours	Views of Binary Trees
1 Hour	Path Sum Problems
1 Hour	Tree Construction Problems

Problems Solved in Class:

- Diameter of Binary Tree
- Binary Tree Right Side View
- Path Sum
- Construct Tree from Traversals

Day 6 – Binary Search Trees

Duration	Topics Covered
2 Hours	BST Insert/Delete/Search
2 Hours	Validate BST + Lowest Common Ancestor
1 Hour	Kth Smallest + Iterators
1 Hour	Advanced BST Questions

Problems Solved in Class:

- Validate Binary Search Tree
- BST Iterator
- Lowest Common Ancestor
- Kth Smallest Element in BST

Day 7 – Heap Fundamentals

Duration	Topics Covered
2 Hours	Min Heap, Max Heap, Heapify
2 Hours	Priority Queue in Java
1 Hour	Heap Sort
1 Hour	Heap Practice Problems

Problems Solved in Class:

- Kth Largest Element
- Heap Sort
- Top K Frequent Elements
- Last Stone Weight

Day 8 – Advanced Heap Problems

Duration	Topics Covered
2 Hours	Sliding Window Maximum
2 Hours	Merge K Sorted Lists
1 Hour	Median Finder
1 Hour	Greedy + Heap Concepts

Problems Solved in Class:

- Sliding Window Maximum
- Find Median from Data Stream
- Connect Ropes
- Merge K Sorted Lists

Day 9 – Graph Fundamentals

Duration	Topics Covered
2 Hours	Graph Representation
2 Hours	BFS + DFS
1 Hour	Connected Components
1 Hour	Grid Based Graph Problems

Problems Solved in Class:

- Number of Islands
- Flood Fill
- Clone Graph
- Rotting Oranges

Day 10 – Graph Algorithms

Duration	Topics Covered
2 Hours	Cycle Detection
2 Hours	Topological Sorting
1 Hour	Directed vs Undirected Graphs
1 Hour	Bipartite Graph

Problems Solved in Class:

- Course Schedule
- Alien Dictionary
- Is Graph Bipartite
- Detect Cycle in Graph

Day 11 – Shortest Path Algorithms

Duration	Topics Covered
2 Hours	Dijkstra Algorithm
2 Hours	Shortest Path Problems
1 Hour	Bellman Ford
1 Hour	Optimization Techniques

Problems Solved in Class:

- Network Delay Time
- Cheapest Flights Within K Stops
- Shortest Path in Binary Matrix
- Path With Minimum Effort

Day 12 – MST + DSU

Duration	Topics Covered
2 Hours	Disjoint Set Union
2 Hours	Kruskal Algorithm
1 Hour	Prim Algorithm
1 Hour	Advanced Graph Questions

Problems Solved in Class:

- Redundant Connection
- Accounts Merge
- Minimum Spanning Tree
- Number of Provinces

Day 13 – Dynamic Programming Basics

Duration	Topics Covered
2 Hours	Memoization vs Tabulation
2 Hours	1D DP Patterns
1 Hour	State Transition Thinking
1 Hour	Optimization Techniques

Problems Solved in Class:

- Climbing Stairs
- House Robber
- Decode Ways
- Min Cost Climbing Stairs

Day 14 – Intermediate Dynamic Programming

Duration	Topics Covered
2 Hours	Knapsack Pattern
2 Hours	Subset Sum + Partition Problems
1 Hour	Coin Change Variations
1 Hour	Pattern Recognition

Problems Solved in Class:

- Coin Change
- Partition Equal Subset Sum
- Target Sum
- Perfect Squares

Day 15 – Advanced Dynamic Programming

Duration	Topics Covered
2 Hours	LCS + LIS
2 Hours	Grid DP + String DP
1 Hour	Palindrome DP
1 Hour	Hard DP Problems

Problems Solved in Class:

- Longest Common Subsequence
- Longest Increasing Subsequence
- Edit Distance
- Burst Balloons

Day 16 – Revision + Mock Interviews

Duration	Topics Covered
2 Hours	Complete DSA Revision
2 Hours	Company Tagged Problems
1 Hour	Mock Coding Round
1 Hour	Contest + Debugging Session

Problems Solved in Class:

- Amazon Tagged Questions
- Google Tagged Questions
- Mixed Graph + DP Problems
- Mock Assessment